Assignment 1

Section 1.1

1

a) `True`

b) `False`

c) `True`

d) `False`

e) Not a proposition

f) Not a proposition

5

a) It is not the case that “Mei has an MP3 player.”

b) It is not the case that “There is no pollution in New Jersey.”

c) It is not that case that `2 + 1 = 3`

d) The summer in Maine is not hot or not sunny.

7

a) It is not the case that “Steve has more than 100 GB free disk space on his laptop.”

b) Zach does not block e-mails from Jennifer or does not block texts from Jennifer.

c) It is not the case that `7 * 11 * 13 = 999`

d) It is not the case that “Diane rode her bicylce 100 miles on Sunday.”

8

Smartphone	RAM	ROM	Camera
A	256MB	32GB	8MP
B	288MB	64GB	4MP
C	128MB	32GB	5MP

a) `true`

b) `true`

c) `false`

d) `false`

e) `false`

9

Companny	Annual Revenue	Net Profit
Acme Computer	\$138b	\$8b
Nadir Software	\$87b	\$5b
Quixote Media	\$111b	\$13b

a) `false`

b) `true`

c) `true`

d) `true`

e) `true`

11h

p: Swimming at the New Jersey shore is allowed

q: Sharks have been spotted near the shore

Expression	Transformation/Reason
$\neg p \wedge (p \vee \neg q)$	
$(\neg p \wedge p) \vee \neg q$	Associative Law
$(\texttt{false}) \vee \neg q$	Negation Law
$\neg q$	Identity Law

It is not the case that sharks have been spotted near the shore.

13

p: It is below freezing

q: It is snowing

a) $p \wedge q$

b) $p \wedge \neg q$

c) $\neg p \land q$

d) $p \lor q$

e) $p \to q$

f) $p \lor q \land (p \to \neg q)$

g) $p \leftrightarrow q$

16d

Variable	Definition
p	You get an A on the final exam
q	You do every exercise in this book
r	You get an A in this class

"You get an A on the final, but you don't do every exercise in this book; neverthless, you get an A in this class." $\equiv p \land \neg q \land r$

17

a) $r \land \neg p$

b) $\neg p \land q \land r$

c) $r \to (q \leftrightarrow p)$

d) $\neg q \neg p \land r$

e) $(\neg r \land \neg p) \to q$

19

a) `false`

b) `true`

c) `true`

d) `true`

25h

"Jan will go swimming unless the water is too cold" ≡ "If the water is too cold, Jan will not go swimming and if the water is not too cold, Jan will go swimming" ≡ "Jan will go swimming if and only if the water is not too cold" $\equiv p \leftrightarrow \neg q$

29a

Converse: If I will ski tomorrow, then it snowed today.

Inverse: If it did not snow today, then I will not ski tomorrow.

Contrapositive: If I will not ski tomorrow, then it did not snow today.

30c

p: I stay up late

q: I sleep until noon

Conditional	$p \to q$	When I stay up late, it is necessary that I sleep until noon.
Converse	$q \to p$	If I sleep until noon, then I stay up late.
Inverse	$\neg p \to \neg q$	If I do not stay up late, then I do not sleep until noon.
Contrapositive	$\neg q \to \neg p$	If I do not sleep until noon, then I do not stay up late.

Because the if-clause and main-clause of the original conditional are both present-tense, it is a real-conditional and therefore the statement has no implied timeframe – it only expresses a general, overarching relationship between the two clauses. So, the tenses of both clauses of the converse, inverse, and contrapositive should be present-tense as well.

32c

Variables: p, r, s, t, u, v

Truth Table Rows = 2^Nvariables = 2^6 = 64

33d

p	q	$(p \lor q)$	$(p \land q)$	$(p \lor q) \to (p \land q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

35f

p	q	$p \oplus q$	$p \oplus \neg q$	$(p \oplus q) \to (p \oplus \neg q)$
T	T	F	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	T	T

37a

p	q	$\neg q$	$p \to \neg q$
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	T

39a

p	q	r	$\neg q$	$\neg q \lor r$	$p \to (\neg q \lor r)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	T	T	T
F	F	F	T	T	T

48a

1 1000 ∧ (0 1011 ∨ 1 1011) ≡ 1 1000 ∧ 1 1011 ≡ **1 1000**

Section 1.2

1

$\neg e \to \neg a$

3

$(r \land \neg m \land \neg b) \to g$

5

$(a \land (b \lor p) \land r) \rightarrow e$

11

p: The router can send packets to the edge system

q: The router supports the new address space

r: The latest software release is installed

1. The router can send packets to the edge system only if it supposrts the new address space

   $p \rightarrow q$

2. For the router to support the new address space, it is necessary that the latest software release is installed

   $q \rightarrow r$

3. The router can send packets to the edge system only if the latest software release is installed

   $p \rightarrow r$

4. The router does not support the new adress space

   $\neg q$

$(p \rightarrow q) \land (q \rightarrow r) \land (p \rightarrow r) \land \neg q$

This is satisfiable when `p` is `true`, `q` is `false`, and `r` is `true`, so the specifications are consistent.

17

The two trunks that do not hold the treasure are empty. To win, you must select the correct trunk.

- Trunk 1 Inscription: The treasure is in Trunk 3
- Trunk 2 Inscription: The treasure is in Trunk 1
- Trunk 3 Inscription: This trunk is empty

A: The treasure is in Trunk 1

B: The treasure is in Trunk 2

C: The treasure is in Trunk 3

The Queen, who never lies says

a) “All the inscriptions are false.”

$\neg C \land \neg A \land C$

The Queen cannot say this

b) “Exactly one of the inscriptions is true.”

$C \oplus (A \oplus \neg C)$

Satisfiable when C is true

c) “Exactly two of the inscriptions are true.”

Satifiable when A is true

d) “All three inscriptions are true.”

$C \land A \land \neg C$

The Queen cannot say this

30

A: A is the knave

B: B is the knave

C: C is the knave

A says “I am the knave” and B says “I am the knave” and C says “I am the knave”

$(A \to \neg A) \land (B \to \neg B) \land (C \to \neg C)$

There are no solutions

31

A says “I am the knight” and B says “A is telling the truth” and C says “I am the spy”

Satisfiable when A is the knight, B is the spy, and C is the knave.

35

A says “I am not the spy” and B says “I am not the spy” and C says “I am not the spy”

The spy has to be lying, so there will be two people lying, implying there are two spies, so there are no solutions.

39

- If the butler is telling the truth, then so is the cook
- the cook and the gardner cannot both be telling the truth
- the gardner and the handyman are not both lying
- if the handyman is telling the truth, then the cook is lying

`A`: The butler is telling the truth

`B`: The cook is telling the truth

`C`: The gardner is telling the truth

`D`: The handyman is telling the truth

$A \rightarrow B$

$\neg B \lor \neg C$

$\neg(\neg C \lor \neg D) \equiv C \land D$

$D \rightarrow \neg B$

Satifiable when `A` is true or false, `B` is false, `C` is true, and `D` is true.